The Collapse of Heterogeneity in Silicon Philosophers

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Question

Can an LLM stand in for a population of minds?

Setup

- 277 PhilPeople philosophers
- 100 PhilPapers 2020 questions
- 7 profile-conditioned LLMs
- Temperature 0; Accept → Reject coded onto $[0, 1]$
- Checked against PhilPapers 2020 ($N = 1{,}785$)

Silicon sampling

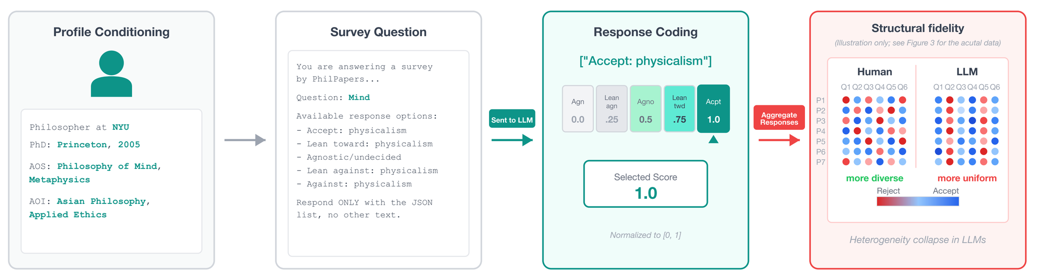

Fig. 1. Profile → question → coded stance → fidelity.

Spread collapses (Table 1)

Source	Resp%	Per-Q Var
Human	19.1	0.062
Claude Sonnet 4.5	36.2	0.043
Llama 3.1 8B	63.0	0.042
Mistral 7B	70.0	0.029
GPT-4o	48.4	0.027
GPT-5.1	49.1	0.026

Humans are 1.4–2.4× more variable than every LLM.

Correlations inflate (Table 2)

Source	Max $\lvert r\rvert$	% Sig.
PhilPapers 2020	0.29	$\sim$5
Human sample	0.57	$\sim$5
Claude Sonnet 4.5	0.82	13.7
GPT-5.1	0.93	13.3

LLMs invent 2–3× more significant specialist links.

A social world model is adequate only if it matches the **distribution of human minds.**

Silicon philosophers **fail this test**: they are **1.4–2.4×** too uniform, and they disagree along the **wrong axes**.

Take a picture for the paper and data explorer.

Human vs. Claude

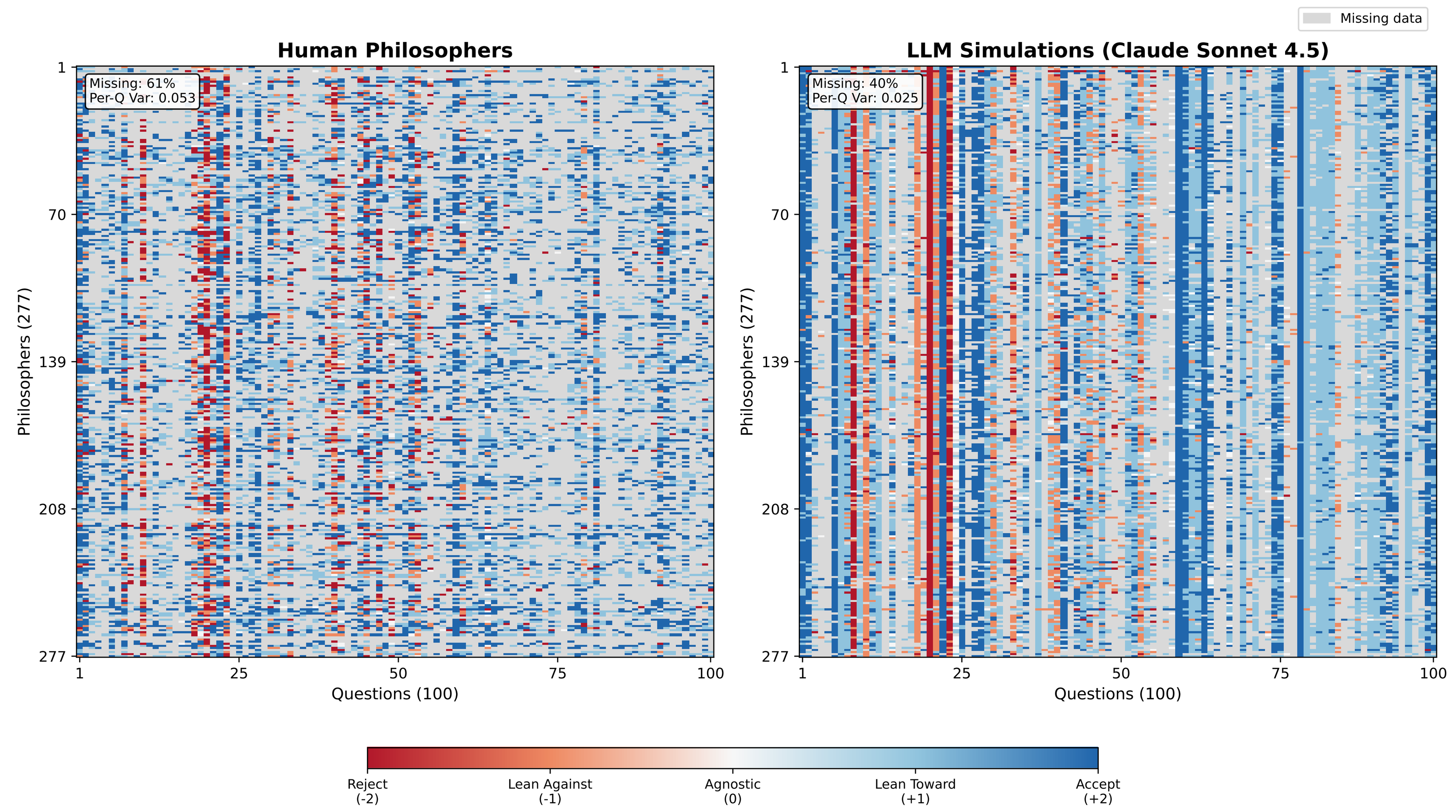

Fig. 3a. Each cell is one philosopher on one question. Humans speckled; Claude forms vertical bands.

Wrong axes (Table 7)

Source	Var(6)	Elem r
Human	22.4%	–
Claude Sonnet 4.5	31.7%	0.108
Llama 3.1 8B (FT)	33.5%	0.079
GPT-4o	28.0%	0.055
GPT-5.1	28.9%	0.049

Models collapse disagreement onto fewer, topical axes. Best pairwise alignment is still weak ($r \leq 0.11$).

Stereotypes fill the gap (Table 3)

Specialist effect	Human	LLM
Biology → biol. identity	+11 pp	+43 pp
Biology → psych. identity	+4 pp	−66 pp
Ancient → Aristotelian	+2 pp	+69 pp

Ground truth n.s.; LLM effects significant in 3–7 of 7 models.

DPO does not restore it (Table 10)

Llama 3.1 8B	Entropy	Corr. KL
Base	0.662	0.155
DPO fine-tuned	0.605	0.062

Structure improves (KL −60%); diversity falls (entropy −8.6%).